

ISHEETA VERMA

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EDUCATION

Binghamton University, State University of New York

Master of Science in Computer Science | GPA: 3.4/4.00

May 2025

Relevant Coursework: Social Media Data Science Pipeline, Design & Analysis of Algorithms, Database Systems, Distributed Systems, Functional PL, Machine Learning

Jaypee Institute of Information Technology Noida, India

Bachelor of Technology in Electronics and Communication

May 2021

TECHNICAL SKILLS

Languages: Python, Java, C++, C, HTML/CSS, Javascript, Typescript, Bash, SQL

Data Stores: MySQL, SQL Server, MongoDB, HDFS, PostgreSQL, Cassandra, OracleDB, Azure Data Lakes, Azure SQL, AWS (S3, Redshift, EC2, RDS, Lambda), GCP (BigQuery, CloudSQL), DynamoDB

Infra and Tools: Docker, Kubernetes, CI/CD (GitHub Actions, GitLab CI), Git, Linux, Unix, Hadoop, MapReduce, Jira, Apache Spark, Databricks, Kafka, Airflow, Power BI, Pandas, Snowflake, Azure Data Factory, Terraform

Domain Knowledge: Data Analysis, Visualization, Microservices, Machine Learning, API Development, System Design, OOPs

Certifications: Microsoft AZ-900 Fundamentals

PROFESSIONAL EXPERIENCE

DXC Technology, Associate Professional Software Engineer

Feb 2021- Aug 2023

- Accelerated release cycles by 50% by integrating **PyTest** and **Apache Airflow** into automated testing workflows, reducing manual QA effort by 20% and enabling faster feature deployment to production
- Designed **Spark/Airflow ETL pipelines** to process **10TB+ monthly data** from AWS S3 to Redshift, improving data processing speed by **40%** and enabling near real-time analytics
- Built a **fault-tolerant Kafka-Flink streaming pipeline** to process multi-terabyte customer interaction data with millisecond latency, powering real-time personalization & insights
- Enhanced API responsiveness by **30%** via optimized **MongoDB queries**, **Redis caching**, and **index tuning**, reducing dashboard load times by **25%** under high traffic
- Reduced incident response time by 40% by deploying scalable **Python/Docker apps on AWS (EC2, RDS, S3)** and integrating **Grafana** and **CloudWatch** to monitor system health, latency, and resource utilization
- Improved **microservices** reliability and throughput by 25% by implementing **gRPC + Kafka** for scalable, low-latency communication
- Enhanced client integration by 35% by designing **RESTful APIs (Python Flask/FastAPI, PostgreSQL)** with 99.99% uptime, lowering partner support tickets by 30%
- Strengthened data security via **role-based access control** in AWS S3/Redshift, improving compliance and client satisfaction by 40%
- Boosted user engagement conversion rates by 12% by integrating **React + TypeScript** dashboards with **GraphQL/REST APIs** and validating flows via **A/B testing**
- Increased reporting speed by 35% with Snowflake fact-dimension schemas using clustering, materialized views for enterprise analytics
- Reduced deployment time by 50% through **GitHub Actions**-based CI/CD automation in agile sprints, enabling fast, error-free releases

Elwave Digitech(I) Private Limited, Software Engineer

June 2020- Jan 2021

- **Engineered** real-time backend services in **Java/Python** using SOLID principles and design patterns or scalable, fault-tolerant microservices for high-availability systems
- **Developed** low-latency, event-driven data pipelines using **TCP/IP protocols** and secure REST APIs for seamless data exchange between on-prem systems, cloud analytics, and CRM platforms
- **Optimized** large-scale data processing by building multi-threaded Java workflows and SQL-based ETL, boosting system performance and throughput by **35%**

PROJECT EXPERIENCE

E-commerce Data Modernization with Snowflake

Dec 2024 – Jan 2025

- Designed and implemented a Snowflake data warehouse, integrating Azure Data Lake to streamline ingestion, cutting processing time from 24 to 1 hour, and enabling real-time analytics for 5M customers via a three-tier medallion architecture
- Leveraged Snowflake's **Streams, Time Travel, and Zero Copy Cloning** to enhance data integrity, automate data transformations, and support advanced analytics and machine learning applications

Medical Imaging Pipeline and Pneumonia Detection Using Deep Learning

Nov 2024 – Dec 2024

- Developed a pipeline for DICOM file preprocessing and visualization using **PyDicom, SimpleITK, and nibabel**, enhancing data quality and optimizing 2D/3D imaging workflows through spatial resampling and orientation standardization, reducing processing time by 30%
- Built a ResNet18 CNN for pneumonia detection with 90% accuracy using preprocessing, normalization, and Class Activation Maps

Sentiment and Engagement Analysis of Social Media Platforms

Oct 2024 – Nov 2024

- Built an analytics dashboard using **Flask, PostgreSQL, and Plotly** to visualize user engagement, sentiment analysis, and toxic content trends across online platforms
- Developed **real-time data pipelines** and advanced relational database queries to track platform-specific metrics, analyze political events' impact, and detect hate speech trends